CS166 Lab4 Assignment
The Relational Algebra & SQL

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CS166 - Spring 2026

April 24, 2026

Assuming the following relations:

```
BOOKS(DocID, Title, Publisher, Year)
STUDENTS(StId, StName, Major, Age)
AUTHORS(AName, Address)
borrows(DocId, StId, Date)
has-writter(DocId, AName)
describes(DocId, Keyword)
```

Write the follwing quiries in relational algebra and the equivalent SQL Queries.

1. List the year and title of each book
   - Relational Algebra: $\pi_{\text{Year, Title}}$ (BOOKS)
   - SQL: `SELECT B.Year, B.Title`
     `FROM BOOKS B`
2. List all information about students whose major is CS
   - Relational Algebra: $\sigma_{\text{Major = 'CS'}}$ (STUDENTS)
   - SQL: `SELECT *`
     `FROM STUDENTS S`
     `WHERE S.Major = 'CS'`
3. List all students with books they can borrow
   - Relational Algebra: STUDENTS $\bowtie$ borrows
   - SQL: `SELECT *`
     `FROM STUDENTS S`
     `JOIN borrows B`
     `WHERE S.StId = B.StId`
4. List all books published by McGraw-Hill before 1990
   - Relational Algebra: $\sigma_{\text{Publisher = McGraw-Hill'}}$ AND $\sigma_{\text{Year < 1990}}$ (BOOKS)
   - SQL: `SELECT *`
     `FROM BOOKS B`
     `WHERE B.Publisher = 'McGraw-Hill' AND B.Year < 1990`
5. List the name of those authors who are living in Davis
   - Relational Algebra: π_{AName} ($\sigma_{\text{Address = 'Davis'}}$ (AUTHORS))
   - SQL: `SELECT A.AName`
     `FROM AUTHORS A`
     `WHERE A.Address = 'Davis'`
6. List the name of students who are older than 30 and who are not studying CS
   - Relational Algebra: π_{StName} ($\sigma_{\text{Age > 30}}$ AND $\sigma_{\text{Major != 'CS'}}$ (STUDENTS))
   - SQL: `SELECT S.StName`
     `FROM STUDENTS S`
     `WHERE S.Age > 30 AND S.Major != 'CS'`

7. Rename AName in the relation AUTHORS to Name
   - Relational Algebra: $\rho_{\text{Name/AName}}$ (AUTHORS)
   - SQL:
     ```
     ALTER TABLE AUTHORS
     RENAME COLUMN AName TO Name
     ```

8. List the names of all students who have borrowed a book and who are CS majors
   - Relational Algebra: π_{StName} ($\sigma_{\text{Major = 'CS'}}$ (STUDENTS) $\bowtie$ borrows)
   - SQL:
     ```
     SELECT DISTINCT S.StName
     FROM STUDENTS S
     JOIN borrows B
     WHERE S.StId = B.StId AND S.Major = 'CS'
     ```

9. List the title of books written by the author "Jones"
   - Relational Algebra: π_{Title} ($\sigma_{\text{AName = 'Jones'}}$ (AUTHORS) $\bowtie$ has-writter $\bowtie$ BOOKS)
   - SQL:
     ```
     SELECT B.Title
     FROM BOOKS B
     JOIN has-writter HW WHERE B.DocID = HW.DocId
     JOIN AUTHORS A WHERE HW.AName = A.AName
     WHERE A.AName = 'Jones'
     ```

10. As previous, but not books that have the keyword "database"
    - Relational Algebra: π_{Title} ($\sigma_{\text{AName = 'Jones'}}$ (AUTHORS) $\bowtie$ has-writter $\bowtie$ BOOKS) / π_{Title} ($\sigma_{\text{Keyword = 'database'}}$ (describes) $\bowtie$ BOOKS)
    - SQL:
      ```
      SELECT B.Title
      FROM BOOKS B
      JOIN has-writter HW ON B.DocID = HW.DocId
      JOIN AUTHORS A ON HW.AName = A.AName
      WHERE A.AName = 'Jones' AND B.DocID NOT IN (SELECT D.DocId
      FROM describes D
      WHERE D.Keyword = 'database')
      ```

11. Find the name of the youngest student
    - Relational Algebra: π_{StName} ($\sigma_{\text{Age = MIN(Age)}}$ (STUDENTS))

    temp = π_{Age} (STUDENTS) / π_{Age} ($\sigma_{\text{Age1 < Age}}$ (STUDENTS $\bowtie$ $\rho_{\text{Age/Age1}}$ (STUDENTS)))

    π_{StName} ($\sigma_{\text{Age = temp}}$ (STUDENTS))
    - SQL:
      ```
      SELECT S.StName
      FROM STUDENTS S
      WHERE S.Age = (SELECT MIN(Age) FROM STUDENTS)
      ```